Lecture 8: Eigenanalysis

Seonyeong Park, Ph.D. (: sp33@illinois.edu)
Research Assistant Professor
Bioengineering
University of Illinois Urbana-Champaign
February 17, 2025

🔑 Learning Objectives

By the end of this lecture, you should be able to understand:

- Eigenvalues and eigenvectors
- Spectral decomposition
- Commutation operator

Eigenvectors and Eigenvalue Spectra

If the **domain** and **range** of a linear operator $\mathcal{A}$ are the **same space**, it may be possible to find vectors ψ such that:

$$\mathcal{A}\psi = \lambda\psi$$
$$(\mathcal{A} - \lambda\mathcal{I})\psi = \mathbf{0}$$

where:

- ψ: Eigenvector
- λ: Eigenvalue (a scalar constant)
- $\mathcal{I}$: Identity operator

A special property of eigenvectors is that under the action of $\mathcal{A}$:

- The length of the vector may change.
- The angle of the vector remains unchanged.

⭐ *Eigenvectors and eigenvalues are **essential tools** in **analyzing imaging systems** and **solving inverse problems**.*

Eigenvalue Spectrum

The set of all eigenvalues of $\mathcal{A}$ forms the **eigenvalue spectrum**. Depending on whether the eigenvalues are discrete or continuous:

$$\mathcal{A}\psi_n = \lambda_n \psi_n \quad (\textbf{Discrete index } n)$$
$$\mathcal{A}\psi_v = \lambda_v \psi_v \quad (\textbf{Continuous index } v)$$

where:

- λ_n: n-th eigenvalue
- $\{\lambda_n\}$: Eigenvalue spectrum of $\mathcal{A}$

Similarity Transformations

If $\mathcal{W}$ is an **invertible opertor**, then applying it to an eigenvalue equation:

$$\mathcal{A}\psi_n = \lambda_n \psi_n$$
$$\mathcal{W}\mathcal{A}\psi_n = \lambda_n \mathcal{W}\psi_n$$
$$\mathcal{W}\mathcal{A}\mathcal{I}\psi_n = \lambda_n \mathcal{W}\psi_n$$
$$\mathcal{W}\mathcal{A}\mathcal{W}^{-1}\mathcal{W}\psi_n = \lambda_n \mathcal{W}\psi_n$$
$$\mathcal{W}\mathcal{A}\mathcal{W}^{-1}\mathcal{W}\psi_n = \lambda_n \mathcal{W}\psi_n$$
$$\mathcal{A}_{new}\psi_{new\,n} = \lambda_n \psi_{new\,n}$$

This transformation is called a **similiarity transformation**, which modifies the operator while preserving its eigenvalue spectrum $\{\lambda_n\}$ by means of an invertible operator $\mathcal{W}$.

The operators $\mathcal{A}$ and $\mathcal{A}_{new} = \mathcal{W}\mathcal{A}\mathcal{W}^{-1}$ are **similar operators** and have the **same eigenvalue spectrum** $\{\lambda_n\}$.

Action of Similar Operators

Let $\mathcal{A}$ and $\mathcal{A}'$ be **similar operators** that share the same eigenvalue spectrum $\{\lambda_n\}$. They are related by an invertible operator $\mathcal{W}$ as follows:

$$\mathcal{A}' = \mathcal{W}\mathcal{A}\mathcal{W}^{-1}.$$

Under this transformation:

$$\mathcal{A}f = g \quad \text{Old basis vectors: } u_n$$
$$\mathcal{A}'f' = g' \quad \text{New basis vectors: } u'_n = \mathcal{W}u_n$$
$$\mathcal{A}\mathcal{W}f = \mathcal{W}g$$

The two equations $\mathcal{A}'f' = g'$ and $\mathcal{A}f = g$ represent the same mathematical relation in terms of different basis vectors.

- This transformation represents a **change of coordinate axes**, where both vectors and operators are expressed in a different basis.
- For a matrix representation of the operator $\mathcal{A}$, by the transformation, its elements change ($A_{mn} \neq A'_{mn}$), but the **effect of the matrix as an operator does not change** if the relevant vectors are also changed appropriately.
- The marix representations depends on the choice of basis, while the underlyaing operator does not.

Eigenanalysis in Finite-dimensional Spaces

For an operator $\mathcal{A}$ acting on an N-dimensional space:

$$\mathcal{A} : \mathbb{E}^N \to \mathbb{E}^N, \mathbf{A} \in \mathbb{E}^{N\times N}.$$

Eigenvectors are column vectors: $\psi_n \in \mathbb{E}^N$.

The **eigenvalue equation** is:

$$(\mathbf{A} - \lambda\mathbf{I})\psi = \mathbf{0}$$

where:

- $\mathbf{I}$: $N \times N$ identity matrix (unit matrix)

If $(\mathbf{A} - \lambda\mathbf{I})$ is **invertible (non-singular)**, the equation has only trivial solutions $\psi = \mathbf{0}$.

Nontrivial solutions (the solutions that are not $\mathbf{0}$) exist only if $(\mathbf{A} - \lambda\mathbf{I})$ is **singular (not invertible)**, i.e., $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$.

The **eigenvalues** can be obtained by solving the equation. Once the eigenvalues are determined, substituting them back into $(\mathbf{A} - \lambda\mathbf{I})\psi = \mathbf{0}$ yields the corresponding **eigenvectors**.

For a square matrix $\mathbf{A}$, the following properties are equivalent:

- $\mathbf{A}$ is **non-singular**.
- $\mathbf{A}$ is **invertiable**.

- $\mathbf{A}$ is **full-rank**.
- $\mathbf{A}$ has **no null space**.
- $\mathbf{A}$ has **no zero eigenvalues**.

Eigenanalysis of Hermitian Operators

If $\mathcal{A}$ is **Hermitian**, it satifies: $\mathcal{A}^\dagger = \mathcal{A}$, and its eigenvalues are real: $\lambda = \lambda^*$.

Using properties of the inner product:

$$\begin{aligned}\mathcal{A}\psi &= \lambda\psi \\ \langle \psi, \mathcal{A}\psi\rangle &= \langle \psi, \lambda\psi\rangle \\ \langle \psi, \mathcal{A}\psi\rangle &= \lambda\langle \psi, \psi\rangle \\ \langle \psi, \mathcal{A}\psi\rangle &= \lambda\|\psi\|^2\end{aligned}$$

Using the definition of the adjoint:

$$\begin{aligned}\langle \psi, \mathcal{A}\psi\rangle &= \langle \mathcal{A}^\dagger\psi, \psi\rangle \\ \langle \psi, \mathcal{A}\psi\rangle &= \langle \mathcal{A}\psi, \psi\rangle \\ \langle \psi, \mathcal{A}\psi\rangle &= \langle \lambda\psi, \psi\rangle \\ \langle \psi, \mathcal{A}\psi\rangle &= \lambda^*\langle \psi, \psi\rangle \\ \langle \psi, \mathcal{A}\psi\rangle &= \lambda^*\|\psi\|^2\end{aligned}$$

🧠 *Recall the definition of the adjoint adjoint operator* $\mathcal{H}^\dagger$ (in Lecture 5):
$\langle \mathbf{g}_2, \mathcal{H}\mathbf{f}_1\rangle_{\mathbb{V}} = \langle \mathcal{H}^\dagger\mathbf{g}_2, \mathbf{f}_1\rangle_{\mathbb{U}}$

Thus, the **eigenvalues of a Hermitian operator are real**: $\lambda = \lambda^*$.

A Special Case of $\mathcal{B}^\dagger\mathcal{B}$

For any arbitrary operator $\mathcal{B}$, the operator $\mathcal{B}^\dagger\mathcal{B}$ is **Hermitian**:

$$(\mathcal{B}^\dagger\mathcal{B})^\dagger = (\mathcal{B})^\dagger(\mathcal{B}^\dagger)^\dagger = \mathcal{B}^\dagger\mathcal{B}.$$

Use properties of the inner product

$$\langle \mathbf{u}, \mathcal{B}^\dagger\mathcal{B}\mathbf{v}\rangle = \mathbf{u}^\dagger(\mathcal{B}^\dagger\mathcal{B}\mathbf{v}) = \mathbf{u}^\dagger\mathcal{B}^\dagger\mathcal{B}\mathbf{v} = (\mathbf{u}^\dagger\mathcal{B}^\dagger)\mathcal{B}\mathbf{v} = (\mathcal{B}\mathbf{u})^\dagger\mathcal{B}\mathbf{v} = \langle \mathcal{B}\mathbf{u}, \mathcal{B}\mathbf{v}\rangle$$

🧠 *Recall the inner product in a discrete form* (in Lecture 2 and Lecture 4):
$\langle \mathbf{f}_1, \mathbf{f}_2\rangle = \sum_{n=1}^{N} f_{1n}^* f_{2n} = \mathbf{f}_1^\dagger\mathbf{f}_2$

Now, consider an eigenvalue equation: $\mathcal{B}^\dagger\mathcal{B}\phi = \mu\phi$.

Taking the inner product with ϕ

$$\langle \phi, \mathcal{B}^\dagger \mathcal{B}\phi \rangle = \langle \phi, \mu\phi \rangle = \mu \|\phi\|^2$$
$$\langle \phi, \mathcal{B}^\dagger \mathcal{B}\phi \rangle = \langle \mathcal{B}\phi, \mathcal{B}\phi \rangle = \|\mathcal{B}\phi\|^2$$
$$\mu \|\phi\|^2 = \|\mathcal{B}\phi\|^2$$
$$\mu = \frac{\|\mathcal{B}\phi\|^2}{\|\phi\|^2}$$

$$\|\mathcal{B}\phi\|^2 \geq 0, \|\phi\|^2 > 0$$
$$\mu \geq 0$$
$$\mu = 0 \,\text{only if}\, \mathcal{B}\phi = \mathbf{0}$$

This shows that $\mathcal{B}^\dagger \mathcal{B}$ is **positive semi-definite** or **non-negative definite** ($\mathcal{B}^\dagger \mathcal{B} \succeq 0$), which means *all eigenvalues of theh operator are non-negative*.

💡 *If an operator $\mathcal{B}^\dagger \mathcal{B}$ has **no null space**, then all of its eigenvalues are strictly positive, meaning the operator is **positive definite** ($\mathcal{B}^\dagger \mathcal{B} \succ 0$).*

💡 *Definiteness of an $N \times N$ **Hermitian matrix** $\mathbf{A}$ (all eigenvalues of $\mathbf{A}$ are real)*

- $\mathbf{A}$ is **positive definite** if and only if all of its eigenvalues are positive ($\mathbf{A} \succ 0$).
- $\mathbf{A}$ is **positive semi-definite** if and only if all of its eigenvalues are non-negative ($\mathbf{A} \succeq 0$).
- $\mathbf{A}$ is **negative definite** if and only if all of its eigenvalues are negative ($\mathbf{A} \prec 0$).
- $\mathbf{A}$ is **negative semi-definite** if and only if all of its eigenvalues are non-positive ($\mathbf{A} \preceq 0$).
- $\mathbf{A}$ is **indefinite** if and only if it has both positive and negative eigenvalues.

Let $\mathcal{A}$ is a **Hermitian** operator satisfying: $\mathcal{A}\psi_1 = \lambda_1 \psi_1$ and $\mathcal{A}\psi_2 = \lambda_2 \psi_2$, where $\lambda_1 \neq \lambda_2$.

Using the definition of the adjoint:

$$\langle \psi_1, \mathcal{A}\psi_2 \rangle = \langle \mathcal{A}^\dagger \psi_1, \psi_2 \rangle$$
$$\langle \psi_1, \mathcal{A}\psi_2 \rangle = \langle \mathcal{A}\psi_1, \psi_2 \rangle$$
$$\langle \psi_1, \mathcal{A}\psi_2 \rangle = \langle \psi_1, \lambda_2 \psi_2 \rangle = \lambda_2 \langle \psi_1, \psi_2 \rangle$$
$$\langle \mathcal{A}\psi_1, \psi_2 \rangle = \langle \lambda_1 \psi_1, \psi_2 \rangle = \lambda_1 \langle \psi_1, \psi_2 \rangle$$

🧠 *Recall the definition of the adjoint adjoint operator $\mathcal{H}^\dagger$* (in Lecture 5):
$\langle \mathbf{g}_2, \mathcal{H}\mathbf{f}_1 \rangle_{\mathbb{V}} = \langle \mathcal{H}^\dagger \mathbf{g}_2, \mathbf{f}_1 \rangle_{\mathbb{U}}$

$$\lambda_2\langle\psi_1,\psi_2\rangle = \lambda_1\langle\psi_1,\psi_2\rangle$$
$$(\lambda_2-\lambda_1)\langle\psi_1,\psi_2\rangle = 0$$
$$\langle\psi_1,\psi_2\rangle = 0$$

Thus, ψ_1 and ψ_2 are **orthogonal**.

Eigenvectors corresponding to different eigenvalues of a *Hermitian operator* are **orthogonal**.

Diagonalization of a Hermitian Operator

Let $\mathbf{A}$be an $N \times N$ Hermitian matrix with a set of **orthonormal eigenvectors** $\{\psi_n\} \in \mathbb{E}^N$ and corresponding real eigenvalues λ_n. The eigenvectors (column vectors) are used to construct the matrix:

$$\mathbf{\Psi} = [\psi_1 \dots \psi_n \dots \psi_N]$$
$$= \begin{bmatrix} \Psi_{11} & \dots & \Psi_{1n} & \dots & \Psi_{1N} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \Psi_{m1} & \dots & \Psi_{mn} & \dots & \Psi_{mN} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \Psi_{N1} & \dots & \Psi_{Nn} & \dots & \Psi_{NN} \end{bmatrix}$$

Element in the m-th row and the n-th column of $\mathbf{\Psi}$

$$= \begin{bmatrix} \psi_{11} & & \psi_{n1} & & \psi_{N1} \\ \vdots & & \vdots & & \vdots \\ \psi_{1m} & \dots & \psi_{nm} & \dots & \psi_{Nm} \\ \vdots & & \vdots & & \vdots \\ \psi_{1N} & & \psi_{nN} & & \psi_{NN} \end{bmatrix}$$

The m-th element of ψ_n

$$[\mathbf{\Psi}^\dagger\mathbf{\Psi}]_{nn'} = \sum_{k=1}^{N} \Psi^*_{kn}\Psi_{kn'} = \sum_{k=1}^{N} \psi^*_{nk}\psi_{n'k}$$
$$= \langle\psi_n,\psi_{n'}\rangle = \delta_{nn'} = \begin{cases} 1, & \text{if } n = n', \\ 0, & \text{if } n \neq n'. \end{cases}$$
$$\mathbf{\Psi}^\dagger\mathbf{\Psi} = \mathbf{I} \quad \text{Completeness (Closure)}$$
$$\mathbf{\Psi}^\dagger = \mathbf{\Psi}^{-1} \quad \text{Unitary}$$

The unitary matrix $\mathbf{\Phi}$ can diagonalize $\mathbf{A}$:

$$[\boldsymbol{\Psi}^\dagger \mathbf{A} \boldsymbol{\Psi}]_{nn'}$$

$$\mathbf{A}\boldsymbol{\Psi}_{n'} = \begin{bmatrix} A_{11} & \dots & A_{1j} & \dots & A_{1N} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ A_{k1} & \dots & A_{kj} & \dots & A_{kN} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ A_{N1} & \dots & A_{Nj} & \dots & A_{NN} \end{bmatrix} \begin{bmatrix} \psi_1 & \dots & \psi_{n'} & \dots & \psi_N \end{bmatrix}$$

$$= \begin{bmatrix} \sum_{j=1}^N A_{1j}\psi_{1j} & \dots & \sum_{j=1}^N A_{kj}\psi_{n'j} & \dots & \sum_{j=1}^N A_{Nj}\psi_{Nj} \end{bmatrix}$$

$$= \sum_{k=1}^N \psi^*_{nk} \sum_{j=1}^N A_{kj}\psi_{n'j}$$

$$\mathbf{A}\phi_{n'} = \lambda_{n'}\phi_{n'} \quad \sum_{j=1}^N A_{kj}\psi_{n'j} = \lambda_{n'}\psi_{n'k}$$

$$= \sum_{k=1}^N \psi^*_{nk}\lambda_{n'}\psi_{n'k}$$

$$= \lambda_{n'} \sum_{k=1}^N \psi^*_{nk}\psi_{n'k} = \lambda_{n'}\langle \psi_n, \psi_{n'} \rangle = \lambda_{n'}\delta_{nn'} = \lambda_n \delta_{nn'}$$

$$\boldsymbol{\Psi}^\dagger \mathbf{A} \boldsymbol{\Psi} = \boldsymbol{\Lambda}$$

where $\boldsymbol{\Lambda}$ is the **diagonal matrix** with the n-th diagonal element equal to λ_n.

Spectral Decomposition of the Hermitian matrix $\mathbf{A}$

Let $\mathbf{A}$ be an $N \times N$ **Hermitian** matrix. From its unitary diagonalization:

$$\boldsymbol{\Psi}^\dagger \mathbf{A} \boldsymbol{\Psi} = \boldsymbol{\Lambda} \quad \text{Unitary matrix } \boldsymbol{\Psi}^\dagger = \boldsymbol{\Psi}^{-1}$$

$$\boldsymbol{\Psi}^{-1} \mathbf{A} \boldsymbol{\Psi} = \boldsymbol{\Lambda}$$

$$\boldsymbol{\Psi}\boldsymbol{\Psi}^{-1} \mathbf{A} \boldsymbol{\Psi} = \boldsymbol{\Psi}\boldsymbol{\Lambda}$$

$$\mathbf{A} \boldsymbol{\Psi} = \boldsymbol{\Psi}\boldsymbol{\Lambda}$$

$$\mathbf{A} \boldsymbol{\Psi}\boldsymbol{\Psi}^{-1} = \boldsymbol{\Psi}\boldsymbol{\Lambda}\boldsymbol{\Psi}^{-1}$$

$$\mathbf{A} = \boldsymbol{\Psi}\boldsymbol{\Lambda}\boldsymbol{\Psi}^{-1}$$

$$\mathbf{A} = \boldsymbol{\Psi}\boldsymbol{\Lambda}\boldsymbol{\Psi}^\dagger$$

$$\mathbf{A} = \sum_{n=1}^N \psi_n \lambda_n \psi_n^\dagger = \sum_{n=1}^N \lambda_n \psi_n \psi_n^\dagger$$

This is the **spectral decomposition** of $\mathbf{A}$, expressing it as a *weighted* sum of **projection operators**, with the *weighting coefficients* just being the **eigenvalues**.

🧠 *Recall that projection operator/projector is an idempotent operator that is also Hermitian* (in Lecture 5): *One specific form of projection operator is* $\mathcal{P} = \mathbf{p}\mathbf{p}^\dagger$, *where* $\mathbf{p}$ *is some vector with unit norm (*$\|\mathbf{p}\|_2 = 1$*).*

The inverse of the Hermitian matrix $\mathbf{A}$ if it exists

$$\mathbf{A} = \sum_{n=1}^{N} \lambda_n \psi_n \psi_n^\dagger$$

$$\text{Orthonormality of } \{\psi_n\}: \ \psi_n^\dagger \psi_m = \langle \psi_n, \psi_m \rangle = \delta_{nm}$$

$$\mathbf{A}^{-1} = \sum_{m=1}^{N} \frac{1}{\lambda_m} \psi_m \psi_m^\dagger$$

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \sum_{n=1}^{N} \lambda_n \psi_n \psi_n^\dagger \sum_{m=1}^{N} \frac{1}{\lambda_m} \psi_m \psi_m^\dagger$$

$$= \sum_{n=1}^{N}\sum_{m=1}^{N} \lambda_n \psi_n \psi_n^\dagger \frac{1}{\lambda_m} \psi_m \psi_m^\dagger = \sum_{n=1}^{N}\sum_{m=1}^{N} \frac{\lambda_n}{\lambda_m} \psi_n \psi_n^\dagger \psi_m \psi_m^\dagger$$

$$= \sum_{n=1}^{N}\sum_{m=1}^{N} \frac{\lambda_n}{\lambda_m} \psi_n \delta_{nm} \psi_m^\dagger$$

$$= \sum_{n=1}^{N} \frac{\lambda_n}{\lambda_n} \psi_n \psi_n^\dagger = \sum_{n=1}^{N} \psi_n \psi_n^\dagger = \mathbf{I}$$

Commuting Matrices

Two Hermitian operators can be **simultaneously diagonalized** by a **unitary transformation** if and only if they commute.

Let $\mathbf{A}$ and $\mathbf{B}$ be commuting Hermitian matrices, i.e., $\mathbf{AB} = \mathbf{BA}$.

$$\mathbf{BA}\psi_n = \mathbf{AB}\psi_n$$

$$\mathbf{A}\psi_n = \lambda_{An}\psi_n$$

$$\mathbf{B}\lambda_{An}\psi_n = \mathbf{AB}\psi_n$$

$$\lambda_{An}\mathbf{B}\psi_n = \mathbf{AB}\psi_n$$

$$\psi_{new\,n} = \mathbf{B}\psi_n$$

$$\lambda_{An}\psi_{new\,n} = \mathbf{A}\psi_{new\,n}$$

This implies that $\psi_{new\,n}$ is also an eigenvector of $\mathbf{A}$ with the same eigenvalue λ_{An}.

There cannot be two linearly independent eigenvectors with the same eigenvalules. $\mathbf{B}\psi_n$ must be a constant times ψ_n, i.e., $\psi_{new\,n} = (\text{Constant})\psi_n$

$$\mathbf{B}\psi_n = \lambda_{Bn}\psi_n$$

Thus, the Hermitian matrices $\mathbf{A}$ and $\mathbf{B}$ share the same eigenvectors, but not necessarily the same eigenvalues.

$$\mathbf{\Psi}^\dagger\mathbf{A}\mathbf{\Psi} = \mathbf{\Lambda}_A \quad \mathbf{\Psi}^\dagger\mathbf{B}\mathbf{\Psi} = \mathbf{\Lambda}_B$$